

TWO HARBORS HELGESON, MN, USA

WMO: **727444**

Lat: **47.049N** Lon: **91.745W** Elev: **329** StdP: **97.43** Time Zone: **-6.00 (NAC)** Period: **94-19** WBAN: **04979**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	**1**	**-27.3**	**-23.9**	**-32.5**	**0.2**	**-26.6**	**-29.2**	**0.3**	**-22.6**	**9.0**	**-5.3**	**8.2**	**-6.3**	**2.4**	**290**	**0.589**	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	**7**	**11.4**	**28.8**	**22.1**	**27.4**	**20.5**	**26.0**	**19.2**	**23.2**	**27.5**	**21.7**	**26.1**	**20.2**	**24.4**	**4.0**	**190**	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	**22.1**	**17.4**	**26.7**	**20.1**	**15.4**	**24.6**	**18.7**	**14.1**	**23.3**	**70.6**	**27.5**	**64.5**	**26.0**	**59.5**	**24.4**	**28.9**	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	**9.0**	**7.9**	**6.9**	DB	**-30.6**	**32.2**	**3.7**	**1.8**	**-33.3**	**33.5**	**-35.5**	**34.5**	**-37.6**	**35.6**	**-40.3**	**36.9**	(4)
(5)				WB	**-30.8**	**25.1**	**3.5**	**1.7**	**-33.3**	**26.3**	**-35.4**	**27.4**	**-37.4**	**28.3**	**-39.9**	**29.6**	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	**Temperatures, Degree-Days and Degree-Hours**		DBAvg	**4.4**	**-11.2**	**-9.4**	**-3.1**	**3.6**	**10.0**	**15.3**	**18.8**	**17.6**	**13.2**	**6.4**	**-1.1**	**-8.0**	(6)
(7)			DBStd	**11.60**	**7.30**	**6.75**	**6.50**	**4.74**	**4.37**	**3.68**	**3.30**	**3.17**	**4.34**	**4.29**	**5.81**	**6.57**	(7)
(8)			HDD10.0	**2899**	**659**	**542**	**406**	**197**	**53**	**4**	**0**	**0**	**18**	**128**	**334**	**557**	(8)
(9)			HDD18.3	**5174**	**917**	**776**	**663**	**441**	**259**	**104**	**35**	**51**	**161**	**369**	**583**	**816**	(9)
(10)			CDD10.0	**866**	**0**	**0**	**1**	**6**	**55**	**162**	**273**	**236**	**114**	**18**	**1**	**0**	(10)
(11)			CDD18.3	**99**	**0**	**0**	**0**	**0**	**2**	**11**	**49**	**29**	**7**	**0**	**0**	**0**	(11)
(12)			CDH23.3	**967**	**0**	**0**	**0**	**2**	**56**	**143**	**453**	**249**	**62**	**2**	**0**	**0**	(12)
(13)			CDH26.7	**202**	**0**	**0**	**0**	**0**	**11**	**24**	**108**	**51**	**8**	**0**	**0**	**0**	(13)
(14)	**Wind**		WSAvg	**2.7**	**2.8**	**2.8**	**2.9**	**3.3**	**3.1**	**2.5**	**2.3**	**2.2**	**2.4**	**2.8**	**2.8**	**2.7**	(14)
(15)	**Precipitation**		PrecAvg	**785**	**28**	**27**	**39**	**67**	**78**	**97**	**96**	**83**	**94**	**84**	**56**	**36**	(15)
(16)			PrecMax	**957**	**73**	**87**	**87**	**213**	**133**	**164**	**242**	**165**	**176**	**173**	**169**	**64**	(16)
(17)			PrecMin	**614**	**4**	**7**	**10**	**16**	**22**	**13**	**28**	**26**	**11**	**28**	**8**	**5**	(17)
(18)			PrecStd	**103**	**17**	**18**	**18**	**40**	**27**	**38**	**53**	**35**	**40**	**36**	**39**	**18**	(18)
(19)	**Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures**	0.4%	DB		**6.2**	**7.9**	**16.0**	**22.2**	**27.9**	**29.3**	**31.6**	**31.1**	**27.7**	**22.4**	**17.1**	**7.1**	(19)
(20)			MCWB		**3.0**	**4.4**	**10.6**	**12.9**	**18.1**	**23.0**	**23.5**	**23.9**	**20.9**	**15.5**	**11.4**	**4.1**	(20)
(21)		2%	DB		**2.6**	**4.1**	**11.3**	**17.7**	**24.1**	**27.1**	**29.0**	**27.7**	**25.1**	**18.8**	**12.2**	**3.9**	(21)
(22)			MCWB		**1.0**	**1.8**	**6.7**	**10.5**	**15.7**	**19.3**	**22.3**	**21.5**	**19.4**	**14.2**	**9.5**	**2.3**	(22)
(23)		5%	DB		**1.0**	**2.4**	**7.8**	**14.8**	**21.4**	**24.9**	**27.6**	**26.2**	**22.7**	**16.5**	**8.9**	**2.5**	(23)
(24)			MCWB		**-0.3**	**0.9**	**4.6**	**8.5**	**14.1**	**18.0**	**21.1**	**19.8**	**17.9**	**12.0**	**6.3**	**1.1**	(24)
(25)		10%	DB		**-1.2**	**0.3**	**5.9**	**12.2**	**18.0**	**22.8**	**26.2**	**24.0**	**21.1**	**13.7**	**7.2**	**1.1**	(25)
(26)			MCWB		**-2.1**	**-0.9**	**2.9**	**6.9**	**12.5**	**17.0**	**19.7**	**18.7**	**16.8**	**10.3**	**4.7**	**-0.1**	(26)
(27)	**Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures**	0.4%	WB		**3.3**	**5.1**	**11.0**	**14.5**	**19.7**	**24.0**	**25.0**	**24.7**	**22.9**	**17.2**	**12.4**	**4.5**	(27)
(28)			MCDB		**6.0**	**7.7**	**14.3**	**19.2**	**26.5**	**28.1**	**29.1**	**29.7**	**26.6**	**19.8**	**15.9**	**6.3**	(28)
(29)		2%	WB		**1.0**	**2.6**	**7.2**	**11.2**	**16.9**	**20.6**	**23.6**	**22.6**	**20.4**	**14.9**	**9.3**	**2.7**	(29)
(30)			MCDB		**2.5**	**4.2**	**9.7**	**17.0**	**22.2**	**25.1**	**27.9**	**27.0**	**24.1**	**18.2**	**12.0**	**3.9**	(30)
(31)		5%	WB		**-0.4**	**0.8**	**4.8**	**9.0**	**15.0**	**19.1**	**22.1**	**20.9**	**18.8**	**12.8**	**6.8**	**0.8**	(31)
(32)			MCDB		**0.8**	**2.2**	**7.8**	**14.4**	**20.1**	**23.4**	**26.7**	**24.8**	**22.2**	**15.9**	**8.7**	**2.2**	(32)
(33)		10%	WB		**-2.3**	**-1.0**	**3.0**	**7.1**	**13.2**	**17.7**	**20.5**	**19.6**	**17.2**	**10.6**	**4.6**	**-0.2**	(33)
(34)			MCDB		**-1.6**	**0.2**	**5.7**	**11.4**	**17.9**	**21.6**	**24.9**	**23.4**	**20.0**	**13.2**	**6.8**	**1.1**	(34)
(35)	**Mean Daily Temperature Range**		MDBR		**9.4**	**10.2**	**9.8**	**10.1**	**11.6**	**11.5**	**11.4**	**11.0**	**10.6**	**9.1**	**8.0**	**8.0**	(35)
(36)		5% DB	MCDBR		**9.2**	**11.0**	**12.5**	**16.3**	**16.9**	**14.8**	**13.5**	**13.2**	**13.4**	**13.9**	**11.5**	**8.2**	(36)
(37)			MCWBR		**7.9**	**8.8**	**9.0**	**10.5**	**9.9**	**9.0**	**8.3**	**8.2**	**9.1**	**9.6**	**8.5**	**6.8**	(37)
(38)		5% WB	MCDBR		**8.8**	**9.6**	**11.9**	**15.4**	**14.9**	**13.1**	**12.9**	**12.1**	**12.1**	**12.5**	**10.6**	**7.6**	(38)
(39)			MCWBR		**7.7**	**7.8**	**8.7**	**10.3**	**9.6**	**9.0**	**8.6**	**8.3**	**8.7**	**9.2**	**8.2**	**6.6**	(39)
(40)	**Clear-Sky Solar Irradiance**		taub		**0.249**	**0.278**	**0.314**	**0.344**	**0.372**	**0.377**	**0.385**	**0.382**	**0.361**	**0.313**	**0.288**	**0.266**	(40)
(41)			taud		**2.424**	**2.352**	**2.359**	**2.384**	**2.369**	**2.386**	**2.390**	**2.421**	**2.450**	**2.560**	**2.521**	**2.431**	(41)
(42)			Ebn at Noon		**864**	**898**	**911**	**911**	**893**	**887**	**874**	**863**	**851**	**853**	**809**	**793**	(42)
(43)			Edh at Noon		**66**	**88**	**103**	**111**	**117**	**116**	**114**	**106**	**95**	**73**	**61**	**59**	(43)
(44)	**All-Sky Solar Radiation**		RadAvg		**1.33**	**2.36**	**3.56**	**4.60**	**5.35**	**5.77**	**6.09**	**5.12**	**3.82**	**2.31**	**1.32**	**0.97**	(44)
(45)			RadStd		**0.14**	**0.23**	**0.33**	**0.47**	**0.39**	**0.40**	**0.25**	**0.35**	**0.24**	**0.28**	**0.16**	**0.10**	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	(46)
(47)	Regional (35 neighbors)	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	(47)

Nomenclature: See separate page